

1. A small firm builds two types of garden sheds.

(a) Unknowns

x = number of Type A sheds produced each day,

y = number of Type B sheds produced each day.

(b) Constraints

Machine time: $2x+3y \leq 30$.

Craftsman time: $5x+5y \leq 60$.

and $x \geq 0, y \geq 0$ and $x, y \in \mathbb{Z}$.

(c) Profit

$P = 60x + 84y$

So, in summary, the linear programming problem is

maximize $P = 60x + 84y$

subject to $2x+3y \leq 30$

$x+y \leq 12$

$x \geq 0, y \geq 0$ and $x, y \in \mathbb{Z}$

2. A farmer has 20 hectares for growing barley and swedes.

(a) Unknowns

x = number of hectares of barley

y = number of hectares of swedes

(b) Constraints

Land $x+y \leq 20$.

Cost $30x+20y \leq 480$.

Manpower $x+2y \leq 36$.

and $x \geq 0, y \geq 0$ and $x, y \in \mathbb{Z}$

(c) Profit

$$P = 100x + 120y$$

So, in summary, the linear programming problem is

$$\text{maximize } P = 100x + 120y$$

$$\text{subject to } x + y \leq 20$$

$$30x + 20y \leq 480$$

$$x + 2y \leq 36$$

$$x \geq 0, y \geq 0 \text{ and } x, y \in \mathbb{Z}$$

X: [Barley(hectares)]

Y: [Swedes (hectares)]

